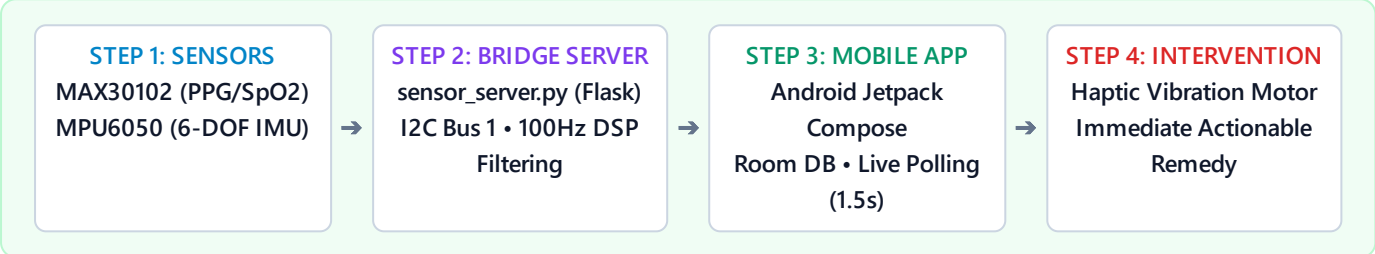
NeuroBand AI™

Complete Execution Guide & 4-Level Sensor Threshold Matrix

Document Version: 2.4 (Production)
Target Stack: Raspberry Pi (I2C) + Android Jetpack Compose
Classification: Engineering & Clinical Protocol

1. System Architecture & Real-Time Flow

NeuroBand AI continuously acquires multi-modal physiological and kinematic telemetry from biometric sensors, applies digital signal processing and machine learning inference, and guides the user via visual alert banners and tactile vibration actuators.



2. Step-by-Step Execution Guide

A. Running the Python Sensor Bridge Server (`sensor_server.py`)

The bridge server exposes a RESTful API at port `5000`. It automatically detects whether physical I2C sensors are connected and seamlessly falls back to testing simulation mode on non-hardware platforms.

1. **Install Dependencies:**

```
pip install flask flask-cors # If running on Raspberry Pi with physical I2C sensors: pip install smbus2
```

2. **Launch the Server:**

```
cd neuroband_server python sensor_server.py
```

The terminal will output:

```
================================================================ NEUROBAND AI - ULTRA PRECISION SENSOR
SERVER Listening on http://0.0.0.0:5000 REST API Endpoint: http://<IP_ADDRESS>:5000/api/biometrics
Simulation Test: http://<IP_ADDRESS>:5000/api/simulate?hr=115
================================================================
```

3. **Browser Dashboard Verification & Quick Threshold Simulation:**
Open `http://localhost:5000` in any web browser. Use the quick buttons to simulate threshold events:
   - `/api/simulate?hr=72&gyro=1.2&spo2=98` (Normal Baseline)
   - `/api/simulate?hr=118` (High Pulse Tachycardia Warning)
   - `/api/simulate?hr=48` (Low Pulse Bradycardia Warning)
   - `/api/simulate?gyro=68` (Tremor & Kinematics Spike)
   - `/api/simulate?spo2=89` (Low SpO2 Hypoxemia Emergency)

B. Running the Android Application in Android Studio

1. Open **Android Studio** and select **Open** ➔ Choose the project folder `Neuroband-AI`.
2. Allow Gradle to sync automatically.
3. Select an active device (Physical Android Phone or Android Emulator API 26+).
4. Click the green **Run ('app')** button (or press `Shift + F10`).
5. Once launched, tap the **Settings** icon (4th tab on bottom bar):
    - If using **Android Emulator** and server is on the same PC: enter host `10.0.2.2`.
    - If using **Physical Phone**: enter the local Wi-Fi IP of your computer or Raspberry Pi (e.g. `192.168.1.50`).
    - Tap **SAVE HOST** and tap **TEST LINK**. The indicator will switch to **ONLINE** 🟢.

C. Switching Light & Dark Themes

The user interface supports both high-contrast Clean Light Theme and Deep Cyber Navy Dark Theme:

- Open the **Settings** tab ➔ In the top **"APPEARANCE & THEME"** card, toggle the switch.
- **Clean Light Theme:** Pristine slate-white surfaces, vibrant cyan & purple accents, and high-contrast dark text designed for bright environments.
- **Dark Cyber Theme:** Deep navy OLED background with glowing cybernetic indicators.

D. Testing Haptic Vibration & Remedies

To verify the physical vibration actuator and on-screen remedy alert:

- Navigate to **Settings** ➔ Under **CONSOLE & ALERTS PREFERENCES**, ensure **"Tactile Vibration Guidance"** is enabled.
- Tap the **"TEST VIBRATION & REMEDY WARNING"** button.
- The phone will vibrate with the warning haptic pattern, switch to the Home tab, and display the alert banner with the clinical remedy.

3. Clinical 4-Level Sensor Threshold Matrix

Every biometric parameter is mapped to 4 actionable levels. When a value leaves Level 2 (Normal), the system immediately engages haptic feedback, displays the warning banner, and instructs the patient with a precise behavioral remedy.

1. Pulse Rate / Heart Rate (BPM) • Optical MAX30102 PPG

Level	Range (BPM)	Clinical Status	App Warning Banner	Vibration Pattern	Actionable Remedy
LEVEL 1	**< 55 BPM**	Bradycardia (Abnormal Low)	⚠️ *"Heart Rate Decreasing (48 BPM) - Low Pulse Warning"*	`200ms ON, 100ms OFF, 200ms ON`	Sit down to prevent dizziness, stay warm, gently flex fingers and wrists, and drink a warm cup of water or tea.
LEVEL 2	**55 – 100 BPM**	**Normal ("It's OK")**	None (Display: *"BPM • OPTIMAL"*)	`No Vibration`	Cardiovascular rhythm is balanced. Continue routine daily activities.
LEVEL 3	**101 – 120 BPM**	Mild Tachycardia (Elevated)	⚠️ *"Heart Rate Increasing (114 BPM) - Elevated Pulse Warning"*	`200ms ON, 100ms OFF, 200ms ON`	Sit down comfortably, drink a glass of cool water, and practice slow 4-7-8 deep breaths to regulate heart rhythm.
LEVEL 4	**> 120 BPM**	Severe Tachycardia (Emergency)	🚨 *"Critical Pulse Alert (135 BPM) - Severe Tachycardia"*	`350ms ON, 150ms OFF x3 (Rapid)`	Halt all physical exertion immediately. Sit upright, sip cool water slowly, apply a cold towel to the neck, and initiate diaphragmatic breathwork.

2. Blood Oxygen Saturation (SpO2 %) • Dual-Wavelength LED Ratio

Level	Range (%)	Clinical Status	App Warning Banner	Vibration Pattern	Actionable Remedy
LEVEL 1	**< 89%**	Severe Hypoxemia (Critical)	🚨 *"Critical SpO2 Alert (87%) - Severe Oxygen Desaturation"*	`350ms ON, 150ms OFF x3 (Rapid)`	**Sit upright, loosen tight collar, practice pursed-lip breathing, and seek immediate medical attention if reading remains below 90%.**
LEVEL 2	**89% – 93%**	Moderate Hypoxia (Warning)	⚠️ *"Blood Oxygen Decreasing (91%) - Low Oxygen Alert"*	`200ms ON, 100ms OFF, 200ms ON`	**Sit upright with relaxed shoulders, open a window for fresh ventilation, and take deep diaphragmatic breaths.**
LEVEL 3	**94% – 95%**	Borderline Acceptable	Informational status pill	`No Vibration`	**Ensure sensor has snug contact with skin. Rest quietly and breathe deeply.**
LEVEL 4	**96% – 100%**	**Normal ("It's OK")**	None (Display: *"SpO2 98%"*)	`No Vibration`	**Blood oxygenation is healthy and optimal.**

3. Tremor & Kinematics (MPU6050 Gyroscope Angular Velocity in °/s & Accel)

Level	Gyro Range (°/s)	Clinical Status	App Warning Banner	Vibration Pattern	Actionable Remedy
LEVEL 1	**< 20 °/s**	**Normal / Resting ("It's OK")**	None (Display: *"Kinematics Stable"*)	`No Vibration`	No involuntary shaking detected. Normal limb motor control.
LEVEL 2	**20 – 45 °/s**	Voluntary Movement / Walking	Tracked as routine pedometer motion	`No Vibration`	Routine functional activity. Movement acceleration is intentional.
LEVEL 3	**46 – 80 °/s**	Elevated Tremor (Parkinsonian Band)	⚠️ *"Tremor & Motion Increasing (58 °/s) - Motor Instability"*	`200ms ON, 100ms OFF, 200ms ON`	Rest forearm firmly on a flat table, relax shoulder muscles, perform slow finger stretching, and check your medication time.
LEVEL 4	**> 80 °/s**	Severe Tremor / Involuntary Spasm	🚨 *"Critical Tremor Alert (92 °/s) - Severe Motor Spasm"*	`350ms ON, 150ms OFF x3 (Rapid)`	Place hands on cushioned surfaces, avoid holding hot drinks or sharp items to prevent drops, and begin guided relaxation breathing.

4. Body / Skin Surface Temperature (°C)

Level	Range (°C)	Clinical Status	App Warning Banner	Vibration Pattern	Actionable Remedy
LEVEL 1	**< 35.5 °C**	Hypothermia Warning (Low)	⚠️ *"Temperature Decreasing (35.1 °C) - Low Temperature"*	`200ms ON, 100ms OFF, 200ms ON`	Put on warm clothing, cover extremities, drink a warm beverage, and remain in a heated environment.
LEVEL 2	**35.5 – 37.2 °C**	**Normal ("It's OK")**	None (Display: *"Temp 36.6 °C"*)	`No Vibration`	Thermoregulation is balanced and stable.
LEVEL 3	**37.3 – 38.3 °C**	Low-Grade Fever (Elevated)	⚠️ *"Temperature Increasing (37.9 °C) - Elevated Temp"*	`Single 200ms Pulse`	Hydrate with cool water, rest in a ventilated area, and apply a cool damp cloth to your forehead or wrists.
LEVEL 4	**> 38.4 °C**	Hyperthermia / High Fever	🚨 *"Critical Temperature Alert (38.8 °C) - High Fever"*	`350ms ON, 150ms OFF x3 (Rapid)`	Hydrate frequently with fluids, wear light breathable clothing, rest in a cool space, and consult physician for antipyretics.

5. Electrodermal Activity (GSR µS) & Autonomic Stress Score

Level	GSR / Score	Clinical Status	App Warning Banner	Vibration Pattern	Actionable Remedy
LEVEL 1	**< 2.0 µS / 0–30**	**Relaxed / Calm ("It's OK")**	None (Display: *"Optimal Autonomic Balance"*)	`No Vibration`	Low sympathetic activation. Excellent mental recovery state.
LEVEL 2	**2.0–3.5 µS / 31–60**	Normal Cognitive Focus	None (Display: *"Active Focus"*)	`No Vibration`	Standard working state during routine tasks.
LEVEL 3	**3.6–5.0 µS / 61–80**	High Cognitive Stress (Warning)	⚠️ *"Stress GSR Surge (4.2 µS) - Cognitive Overload"*	`Single 200ms Guidance Pulse`	Take a 2-minute mindful pause, close your eyes, drink a cup of water, and un-clench your jaw and shoulders.
LEVEL 4	**> 5.0 µS / 81–100**	Acute Burnout Spike (Emergency)	🚨 *"Acute Stress Surge - High Burnout Risk"*	`350ms ON, 150ms OFF x3 (Rapid)`	Disconnect from digital screens immediately. Step outside for fresh air, drink water, and start the app's guided 3-minute Breathing Flow.

4. Explainable AI (SHAP / XAI) Interpretation

In NeuroBand AI, alerts are not opaque labels. Predictions are decomposed into additive feature contributions using **SHAP (SHapley Additive exPlanations)**:

SHAP Additive Attribution Formula:

$$f(x) = E[f(x)] + \Phi_{HeartRate} + \Phi_{SpO2} + \Phi_{Tremor} + \Phi_{Temperature} + \Phi_{GSR}$$

- **Base Value E[f(x)] = 0.05:** The expected baseline risk score when all sensors are inside Level 2 (Normal).
- **Negative Contribution ($\Phi_i \leq 0$):** When a vital is in the normal range, its SHAP score suppresses false alerts ("*Pulse is optimal, reducing overall risk*").
- **Positive Contribution ($\Phi_i > 0$):** When a sensor deviates above or below threshold, its SHAP value increases proportionally, explicitly highlighting the exact biometric responsible for the warning to the clinician and patient.

5. Hardware Pinout Reference (Raspberry Pi 40-Pin Header)

Sensor Pin	Raspberry Pi Pin	Pin Name / Type	Function
MAX30102 VCC	Pin 1	3.3V Power	Sensor Power Supply
MAX30102 GND	Pin 6	Ground	Power Ground
MAX30102 SDA	Pin 3	GPIO 2 (I2C1 SDA)	I2C Data Line (Address: 0x57)
MAX30102 SCL	Pin 5	GPIO 3 (I2C1 SCL)	I2C Clock Line
MPU6050 VCC	Pin 17	3.3V Power	IMU Power Supply
MPU6050 GND	Pin 9	Ground	Power Ground
MPU6050 SDA	Pin 3 (Shared)	GPIO 2 (I2C1 SDA)	I2C Data Line (Address: 0x68)
MPU6050 SCL	Pin 5 (Shared)	GPIO 3 (I2C1 SCL)	I2C Clock Line